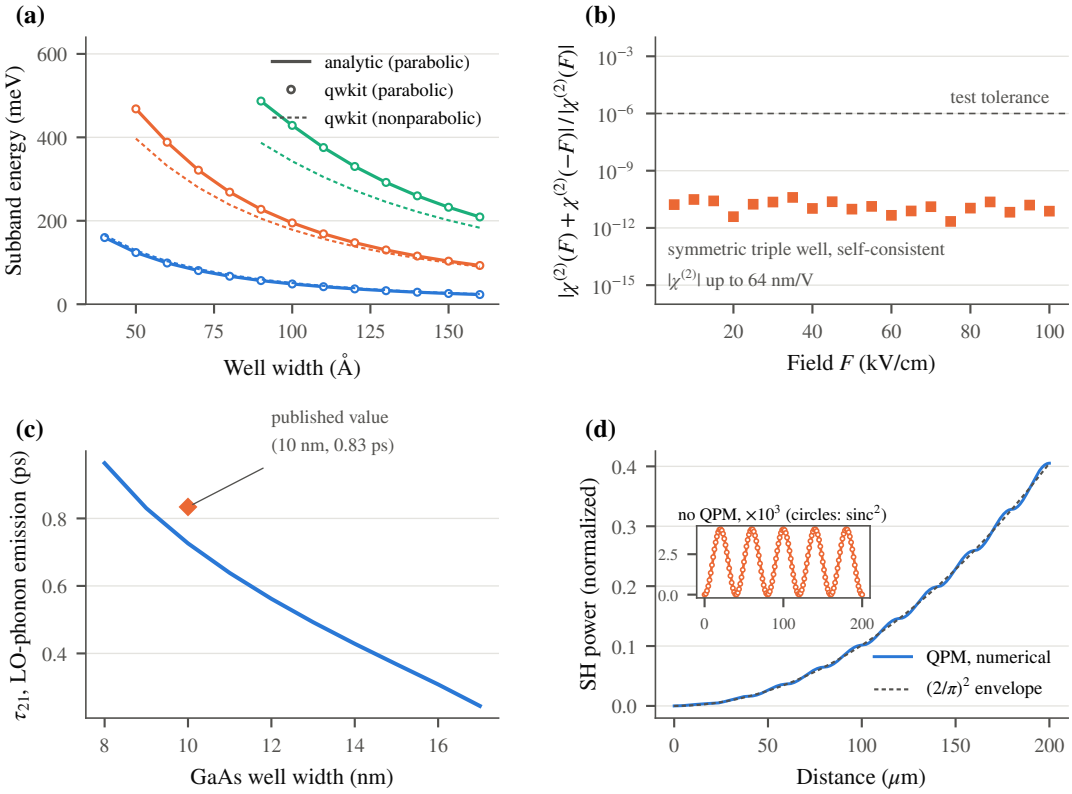
(a)
Subband energy (meV)
600
400
200
0
50
75
100
125
150
Well width (Å)
analytic (parabolic)
qwkit (parabolic)
qwkit (nonparabolic)
(b)
$|\chi^{(2)}(F) + \chi^{(2)}(-F)| / |\chi^{(2)}(F)|$
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
test tolerance
symmetric triple well, self-consistent
$|\chi^{(2)}|$ up to 64 nm/V
20
40
60
80
100
Field F (kV/cm)
(c)
published value
(10 nm, 0.83 ps)
τ_{21}, LO-phonon emission (ps)
0.8
0.6
0.4
8
10
12
14
16
GaAs well width (nm)
(d)
SH power (normalized)
0.4
0.3
0.2
0.1
0.0
no QPM, $\times 10^3$ (circles: sinc^2)
2.5
0.0
0
100
200
QPM, numerical
$(2/\pi)^2$ envelope
0
50
100
150
200
Distance (μm)